INCH-POUND
MIL-DTL-24784/4A (SH)
15 March, 1999
SUPERSEDING
MIL-M-24784/4 (SH)
6 September 1994

ASSOCIATED DETAIL SPECIFICATION

COMMERCIAL OFF-THE-SHELF (COTS) EQUIPMENT MANUAL REQUIREMENTS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the review of commercial off-the-shelf (COTS) manuals (see 6.4.1) and establishes requirements for supplemental data for COTS manuals. This specification is intended as a guide for reviewing personnel to determine the adequacy of existing COTS manuals, and for specifying additional data required to supplement an existing COTS manual. The acceptability of a COTS manual and the need for supplemental data is dependent upon the Governments usage philosophy for each particular item of equipment.

1.2 Application This specification is applicable to manuals that support equipment designed and manufactured for commercial use. It applies to manuals that support commercial equipment used in a commercial or military environment. This specification is not applicable for acquisition of manuals that support developmental items.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 03Q, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-DTL-24784 Manuals, Technical: General Acquisition and Development Requirements.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

DEFENSE LOGISTICS SERVICES CENTER (DLSC)

Cataloging Handbook H4/H8 - Commercial and Government Entity (CAGE).

(Application for copies should be addressed to the Defense Logistics Services Center (ATTN: DLSC-SBB), Federal Center, Battle Creek. MI 49017-3084.)

REGULATIONS

CODE OF FEDERAL REGULATIONS (CFR)
Title 29 CFR - Labor.

EXECUTIVE ORDERS
Executive Order 12196 Occupational Safety and Health Programs for Federal Employees.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402).

2.3 Order of precedence. In the event of a conflict between the text of this document and the

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references cited herein (except for related associated detail specifications), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-DTL-24784.

3.2 Development. A copy of the existing COTS manual shall be required in accordance with the technical manual contract requirements (TMCR) for review by the Government [see Manuals Evaluation (Appendix A) and Acceptability Certification (Appendix B)] to determine whether or not the manual proves acceptable or non-acceptable (see 6.2). An acceptable commercial manual shall contain the data listed in the following subparagraphs, arranged in a sequence that provides continuity. If a manual covers different models, series, or configurations of the equipment, the differences shall be clarified by the use of different data sheets, errata sheets, change sheets, or be included in the text or table(s) in the manual. The manual shall precisely reflect the hardware configuration of the equipment or system acquired by the Government. The amount of data required shall be determined by the complexity or simplicity of the equipment and the usage philosophy of the Government for a particular item of equipment. Usage philosophy includes installation plans, maintenance concept, operational environment, and sparing support approach. The manual shall consist of all data (volumes, folders, inserts, specification sheets, and other documents) required for operation and maintenance of the equipment or system as determined by the Government.

3.2.1 Style and format. The existing COTS manual shall meet the following requirements:

3.2.1.1 Comprehensibility and readability. The text and text-graphics combination of the manual shall be at the level of understanding of the intended user. The reading grade level shall be that of the intended user.

3.2.1.2 Arrangement. The overall arrangement of content, even though vendor unique, shall be organized in a manner sufficient to provide continuity of descriptions, processes, and procedures.

3.2.1.3 Legibility. The entire publication shall be clear and easily read. Text shall be no smaller than 8-point (1 point = 1/72 inch). Parts lists shall be no smaller than 6-point. Lettering on diagrams shall be no smaller than 6-point.

3.2.1.4 Size. Manuals size shall be no smaller than 4 by 5 1/2 inches and no larger than 17 by 11 inches.

3.2.1.5 Safety. The manual shall include safety information that is consistent with the safety standards established by 29 CFR , Part 1910 and Executive Order 12196.

3.2.1.6 Dangers, warnings, cautions, notes. The manual shall contain dangers, warnings,

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cautions, and notes similar to those specified in MIL-DTL-24784.

3.2.1.7 Federal protection standards. Information concerning the use, transportation, handling, storage or disposal of fuel, toxic or hazardous substances, chemicals, ordnance, and munitions shall be clearly stated and shall be consistent with Federal Protection Standards.

3.2.1.8 Illustrations. The manual shall contain illustrations similar to those specified in MIL-DTL-24784 to support the text. The illustrations shall be integrated with the text or their location shall be referenced in the text.

3.2.1.9 Maintenance. The manual shall reflect the maintenance philosophy of the hardware as established by the manufacturer, and the maintenance philosophy shall be such that it can be applied by the intended Government user.

3.3 Content requirements. Except as determined otherwise by the Government, acceptable COTS manuals shall contain, but not be limited to the following:

a. Front matter: Cover or title page, table of contents, and safety precautions.
b. Introduction.
c. Preparation for use.
d. Installation instructions.
e. Principles of operation (at the level required to support maintenance).
f. Operating instructions.
g. Maintenance and servicing instructions (preventive and corrective).
h. Reprogramming.
i. Preparation for shipment.
j. Storage.
k. Parts list.
l. Illustrations and diagrams.
m. Overhaul instructions (as applicable).

3.3.1 Front matter.

3.3.1.1 Cover title page. The manual shall have a cover, title page, or first page showing the date the manual was issued, revision designator (if applicable), the manufacturer's identification name and address, the equipment name, the manufacturer's model designation, and serial or identification numbers for the equipment covered. See Federal Acquisition Regulations concerning contract numbers and markings on publications.

3.3.1.2 Table of contents. A manual containing more than 25 pages shall include a table of contents. It shall list all parts, chapters, sections, and paragraphs in the order or presentation and with the same title used in the text. It shall include a list of illustrations and a list of tables if the manual contains more than 10 illustrations or tables.

3.3.1.3 Safety precautions. The manual shall contain safety precautions where hazards, such as

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high voltage, may be present during installation, operation, or maintenance of the equipment. Health hazards, such as electrostatic discharge, radio frequency radiation, radioactive materials, the presence of poisonous fumes or explosive gases, and the depletion of oxygen from the air in a closed environment, shall be preceded by a danger, warning or caution at the point in the procedure where the hazard is likely to be encountered. See MIL-DTL-24784 for similar requirements.

3.3.2 Introduction. The manual shall contain an introduction, or introductory material, containing the following:

a. Purpose and functions.
b. Capabilities.
c. Performance characteristics.
d. Description (equipment model, dimensions, weight, volume, and center-of-gravity, when applicable).
e. Power and utility information.
f. Environmental information.
g. List of items furnished.
h. List of additional items required for operation and maintenance, but not supplied with equipment.
i. Tools and test equipment.
j. Warranty information.
k. Shipping and handling precautions.
l. Storage data.

3.3.3 Preparation for use. The manual shall contain unpacking and assembling instructions. Inspection for in-shipment damage and instructions on how to handle damaged equipment shall be included.

3.3.4 Installation instructions. Special installation instructions shall include requirements for foundations, ventilation, clearances, plumbing and electrical connections, mountings, wiring runs, initial lubrication, and alignment as applicable.

3.3.5 Principles of operation. The manual shall contain principles of operation information at the technical level necessary for the intended user.

3.3.6 Operating instructions. Operating instructions shall include:

a. Illustrations and explanations of the uses and functions of all controls and indicators (including settings and readings, if applicable).
b. Initial adjustments and control settings.
c. Start up procedures.
d. Normal operation.
e. Operation under emergency, adverse, or abnormal conditions (if different from normal operating procedures).

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f. Shut down procedures.
g. Emergency shutdowns (if required).
h. Interface instructions.

3.3.7 Maintenance and servicing instructions (preventive and corrective). Instructions shall include a list of test equipment, special tools, and materials needed for maintenance and service. This list shall include nomenclature, part and model number, application, range, and accuracy. Instructions or illustrations shall explain or show how test connections are made. Actions and normal indications shall be defined for each test.

3.3.7.1 Cleaning and lubrication. Periodic cleaning and lubrication information, covering types of cleaning agents or lubricants (commercial or military) shall be included. Frequency of lubrications and intervals (monthly, quarterly, and semiannually, hours of operation, mileage, and so forth) shall be presented. Lubrication points and required amounts shall be identified. When applicable pictorial format for lubrication instructions shall be included to clearly illustrate location of each lubrication point. Cleaning and lubrication during repair, replacement, or reassembly shall be included in the instructions covering those actions (see 3.3.7.5).

3.3.7.2 Performance verification. Instructions for calibration of the Test, Measurement, and Diagnostic Equipment needed to restore the equipment to serviceable accuracy (performance verification) shall be provided. These instructions shall be complete step-by-step procedures which will enable the user to check the accuracy of the indications or readings. The location of test connections and the values expected at these points shall be included. Adjustment of built-in self test features shall be included when applicable. Data shall include the recommended frequency of adjustments and verification checks required. When appropriate, the data shall allow for the accomplishment of these adjustments by utilizing a wiring harness, test points, and so forth, that are accessible from outside the equipment case. Data shall include a list of the additional equipment, such as temperature, vacuum, pressure, hydraulic or pneumatic gauges, decade boxes, voltmeters, signal generators, required to accomplish the verification. The listing shall show methods of use or application, range of scales, and specific minimum tolerances or percentages of accuracy.

3.3.7.3 Inspection. Instructions, including scheduling, for inspection of equipment for damage and wear shall be provided with emphasis on allowable service limits such as wear, backlash, end play, balance, and length and depth of scoring. Allowable service limits are acceptable wear tolerances which will not impair performance. They are not to be confused with manufacturing tolerances.

3.3.7.4 Troubleshooting. Malfunctions that might occur during operation of the equipment shall be identified. Troubleshooting data, and fault isolation techniques shall include:

a. The indication or symptom of trouble.
b. The instructions necessary, including test hookups, to determine the cause.
c. Action or reference to action to restore the equipment to operating conditions.

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Applicable information shall be in chart form, logic tree form, in tabular format with appropriate headings, or as logic, block, or schematic diagrams. The established maintenance concept for the associated equipment shall dictate the depth of the troubleshooting such as, localizing a defective circuit card assembly and ending the action by removing and replacing the circuit card assembly as contrasted with the carrying out the troubleshooting process on the defective circuit card assembly to determine the specific piece part that is defective.

3.3.7.5 Disassembly, repair, replacement, and reassembly. Sequential instructions for disassembling, repairing, replacing, and reassembling the equipment shall be provided. Test, adjustment, and check-out data after reassembly shall be provided. Illustrations, including exploded views, shall be used as necessary to support each of these functions.

3.3.8 Reprogramming. A description of reprogrammable memory, reprogramming theory, program setup, program confidence check-out, program loading, and programming procedures shall be provided for equipment having a reprogrammable memory.

3.3.9 Preparation for shipment. Manuals shall contain instructions for the following, as applicable:

a. Disassembly, removal, and separate packaging of electrostatic sensitive discharge devices or fragile components.
b. Use of reusable shipping cases or containers.
c. Special cradles.
d. Mounting.
e. Securing.
f. Covering and preservation.
g. Precautions for shipment.
h. Shipment and unloading.

3.3.10 Storage. Manuals shall contain instructions for the following:

a. Indoor and outdoor storage.
b. Temperature and other environmental limitations.
c. Storage facilities.
d. Dunnage.
e. Ventilation.
f. Revetting.
g. Drainage.
h. Staking.
i. Grounding.
j. Covering.
k. Preservation.

3.3.11 Parts list. The manual shall include a parts list containing positive identification of parts

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for support of the equipment and shall include actual manufacturer or vendor part numbers or generic description necessary to obtain replacement parts.

3.3.11.1 Parts list illustrations. Clear and legible illustrations shall identify component parts and parts relationships.

3.3.11.2 Parts listings. Part numbers and part names shall be shown on illustrations or separately listed. When the illustrations do not contain part numbers and part names, both the illustrations and separate listings shall show either index, reference, or key numbers which will cross-reference illustrated parts to listed parts. The parts list shall cite the actual manufacturer or vendor and their part number or generic description. Parts in the listings shall be grouped by assemblies, subassemblies, and modules with the parts identified to the assembly from which they are components.

3.3.11.3 Common commercial parts. Common commercial hardware and items that are not of special design, such as bolts, washers, nuts, screws, fittings, keys, hinges, wire, cable, gasket material, tubing, hose, and so forth, and are obtain- able from a wide range of sources, shall be identified by part number or the notation "Commercial" instead of a part number. The part name (nomenclature or description) shall be complete enough to facilitate substitution of equivalent items from existing Government stock as in Table I.

TABLE I. Examples of part number descriptions.

Figure and Index No.	Part No.	Part Name (Nomenclature or Description)
2-4	Commercial	NUT, Hex head, plain steel 1/4-20 UNC-3B
2-5	Commercial	WIRE, Electrical, copper, tin plated, No. 14 AWG, 19 strands of No. 27 AWG, 0.250 in. dia.

3.3.12 Operational and maintenance illustrations. Manuals shall contain illustrations (line drawings, photographs or halftones) for locating and identifying all components significant to operation and maintenance and to show configuration and parts relationship for removal and disassembly procedures. Photographs (prescreened) shall be detailed and sharp, free of heavy shadows, distorted objects, cluttered foregrounds or backgrounds, and give good contrast from white, middle tones, and black. Free hand sketches shall not be acceptable.

3.3.12.1 Diagrams. As applicable, the following diagrams shall be included in the manual:

a. Simplified functional block.
b. Locator.
c. Piping.
d. Plumbing.
e. Hydraulic.
f. Schematic.

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g. Electrical.
h. Digital.
i. Other, as necessary.

3.3.12.2 Symbols. Symbols used on illustrations and diagrams shall be standard or common to the trade or commodity. Where nonstandard symbols are used, explanations shall be provided.

3.3.13 Overhaul instructions. When applicable, the manual shall include overhaul instructions to return the equipment to mission capable condition. As a minimum, the instructions shall include the following:

a. List of support equipment, special tools, and facilities required.
b. List of mandatory parts.
c. Preshop analysis, as applicable.
d. Step-by-step procedures for performing all functions including disassembly, removing, replacing, diagnosing, installing, repairing, assembly, in-process testing, adjusting, and inspecting.
e. Final tests to ensure satisfactory performance of the equipment or system overhauled.

3.4 Supplemental data. After the Government determines that the COTS manual meets the acceptability criteria of 3.2 and 3.3, and that supplemental data (see 6.4.2) is required, supplemental data shall be developed in accordance with the TMCR (see 6.2) in contractor format and in accordance with the requirements on the Content/Format Selection Summary Sheet (Appendix C) or other applicable contractual document. This data shall supplement the basic COTS manual and shall not substantially duplicate any data contained in the existing COTS manual.

3.4.1 Identifying technical publication sheet (ITPS). Unless otherwise specified in the TMCR (see 6.2), an Identification Sheet for Commercial Manuals (examples are shown on figures 4-1 and 4-2) shall be tailored to reflect only that information applicable to the acquisition. The appropriate distribution statement, disclosure notice, and authority notice shall appear in accordance with MIL-DTL-24784, as specified by the Government. Other pertinent data shall be inserted by the contractor as provided by the Government.

3.4.2 Style, format, and form.

3.4.2.1 Style and format. Unless otherwise specified in the TMCR (see 6.2), supplemental data shall be in contractor style and format.

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PUBLICATION NUMBER
DATE

IDENTIFYING TECHNICAL PUBLICATION SHEET

1. IDENTIFICATION DATA

2. PURPOSE: This technical publication is issued for the purpose of identifying an authorized commercial manual for Navy use and for providing supplemental technical information thereto.

I. MANUFACTURER: (Name and address)
II. PURCHASE ORDER OR CONTRACT NO.
III. EQUIPMENT: (Type, model, part number, nomenclature, Federal Item Name, serial numbers)
IV. REQUISITION NO: (If furnished by the contracting activity)
V. FEDERAL STOCK NO: (If furnished by the contracting activity)
VI. TITLE: (Operating Instructions, Maintenance Instructions, Parts List, as appropriate)
VII. ADDITIONAL IDENTIFICATION (if any): Volumes, Parts, and so forth
VIII. DATE: (If any)
IX. Preparing activity (if other than prime contractor)
X. Applicable TMCR number
XI. Extent of proposed supplement (10 percent maximum)
    A. Estimated number of proposed text pages.
    B. Estimated number of proposed illustrations.
XII. List identical manual (if any) acquired under another contract.

3. ADDITIONAL COPIES: Additional copies are available from _____. (Contracting activity will furnish information, but if no information is furnished, this paragraph will be omitted and the following paragraph moved up into its place.)

4. FILE LOCATION: The above described commercial manual is filed in ___ ___ ___ ___ ___ ___ ___ ___ ___ ___. (Each library will fill this in if this identifying technical publication sheet is filed separately from the commercial manual.)

FIGURE 4-1. Identification sheet for a commercial manual. (Sample)

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5. COVER PAGE. The cover page shall contain the following information:

a. PUBLISHED BY DIRECTION OF THE COMMANDER, NAVAL SEA SYSTEMS COMMAND

b. For unclassified manuals the following paragraph shall be used:

THIS PUBLICATION IS REQUIRED FOR OFFICIAL USE OR FOR ADMINISTRATIVE OR OPERATIONAL PURPOSES. DISTRIBUTION IS LIMITED TO US GOVERNMENT AGENCIES. OTHER REQUESTS FOR THIS DOCUMENT MUST BE REFERRED TO: COMMANDER, NAVAL SEA SYSTEMS COMMAND (SEA 9941), ARLINGTON, VA 22242

c. For classified manuals the following paragraph shall be used:

NOTICE: THIS MATERIAL CONTAINS INFORMATION AFFECTING THE NATIONAL DEFENSE OF THE UNITED STATES WITHIN THE MEANING OF THE ESPIONAGE LAWS, TITLE 18, USC SECTION 793 AND 794, THE TRANSMISSION OR REVELATION OF WHICH IN ANY MANNER TO AN UNAUTHORIZED PERSON IS PROHIBITED BY LAW.

FIGURE 4-1. Identification sheet for a commercial manual Continued. (Sample)

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PUBLICATION NUMBER
DATE

IDENTIFYING TECHNICAL PUBLICATION SHEET

1. IDENTIFICATION DATA

2. PURPOSE: This technical publication is issued for the purpose of identifying an authorized commercial manual for Navy use and for providing supplemental technical information thereto.

I. MANUFACTURER: (Name and address)
II. PURCHASE ORDER OR CONTRACT NO.
III. EQUIPMENT: (Type, model, part number, nomenclature, Federal Item Name, serial numbers)
IV. REQUISITION NO: (If furnished by the contracting activity)
V. FEDERAL STOCK NO: (If furnished by the contracting activity)
VI. TITLE: (Operating Instructions, Maintenance Instructions, Parts List, as appropriate)
VII. ADDITIONAL IDENTIFICATION (if any): Volumes, Parts, and so forth
VIII. DATE: (If any)
IX. Preparing activity (if other than prime contractor)
X. Applicable TMCR number
XI. Extent of proposed supplement (10 percent maximum)

A. Estimated number of proposed text pages.
B. Estimated number of proposed illustrations.

XII. List identical manual (if any) acquired under another contract.

3. ADDITIONAL COPIES: Additional copies are available from ____ ____ ___ ___ ___.
(Contracting activity will furnish information, but if no information is furnished, this paragraph will be omitted and the following paragraph moved up into its place.)

4. SUPPLEMENTARY DATA

I. (LIST OF EFFECTED PAGES IN BASIC MANUAL. (This will be a summary of the page numbers changed by additions or deletions by the information that follows.)

II. SUPPLEMENTARY INFORMATION. The information contained in the above identified commercial manual is supplemented as follows:

A. Introduction and Description
B. Preparation for use and installation
C. General theory of operation
D. Operating instructions
E. Maintenance instructions
F. Parts list

FIGURE 4-2. Identification sheet and supplemental data provisions for a commercial manual. (Sample)

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3.4.2.2 Form. Supplemental data shall be a separately numbered and bound manual (except as specified in 3.4.2.3) and shall be in one or more of the following forms as specified in the TMCR (see 6.2) by the Government:

a. Review draft copy.
b. Preliminary technical manual.
c. Final reproducible copy.
d. Photo-lithographic negatives
e. Digitized media.
f. Direct image

3.4.2.3 Insert pages. If supplemental pages are less than 50 percent of the existing manual and can be inserted into the existing COTS manual (that is, the manual is not permanently bound), supplemental data may be developed in the TMCR (see 6.2) as change pages (insert pages, appendix pages, and so forth) and issued as a change package.

3.4.3 Content. Content of the supplemental data shall be as specified in the TMCR (see 6.2) on the applicable Content/Format Selection Summary Sheet (Appendix C) or other contractual document provided by the Government. Unless otherwise specified in the TMCR (see 6.2), the data shall be presented in accordance with the requirements specified and in the following order:

a. Cover and title page (see 3.3.1.1).
b. List of effective pages (see 3.4.10).
c. Safety precautions.
d. Table of contents.
e. Reporting of errors statement (see MIL-DTL-24784).
f. Maintenance forms and records.
g. Administrative storage.
h. Destruction of military material to prevent enemy use.
i. Lubrication instructions (see 3.4.11).
j. Preventive maintenance checks and services.
k. Spare and repair parts (see 3.4.12).
l. Recommended changes, user activity comment sheet, or technical manual deficiency report applicable to the particular service, as provided by the Government.
m. Other requirements as specified (when using manuals evaluation checklist as a guide, other required data may be indicated thereon).

3.4.4 Copyrights and proprietary information credit line. Technical manuals shall not contain copyrighted material except as specified in the Federal Acquisition Regulations and Defense Federal Acquisition Regulation Supplement. When copyrighted material is to be included in a technical publication, the developer shall obtain prior written permission from the copyright owner or authorized agent for its use (see 6.2). The signed, written permission shall be delivered together with the final reproducible copy. The written permission shall contain a statement declaring whether or not a copyright credit line is required. When it is necessary to include

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copyright and proprietary material, it shall be clearly identified and the following warning statement shall be included on the title page:

> "This document contains copyright or proprietary materials. Infringement of copyright or proprietary material may violate existing Federal laws and statutes and result in criminal penalties, imprisonment, or removal from office."

3.4.5 Warranty information. If applicable, the COTS manual supplemental data shall contain warranty information pertinent to the equipment covered. It shall include data such as duration of warranty and serial numbers of equipment covered. If warranty is covered separately or in another available document, reference shall be made to that document.

3.4.6 Advertising. Except for the identity of the manufacturer or contractor, supplemental data shall not contain advertising.

3.4.7 Equipment and model coverage. Only equipment and models, accessories, and components specified shall be covered in the supplemental data.

3.4.8 End item identification. Unless otherwise specified in the TMCR (see 6.2), the federal item name, national stock number (NSN), part number, model number, and applicable contract number shall be overprinted on the cover or the first page of the manual.

3.4.9 Publication number and date. The publication number, assigned by the Government, and date shall be overprinted on the cover and first page of the supplement in accordance with MIL-DTL-24784 or as specified in the TMCR (see 6.2).

3.4.10 List of effective pages. A list of effective pages that will include the basic COTS manual and the supplemental data, shall be in accordance with MIL-DTL-24784.

3.4.11 Lubrication instructions. If provided, separate lubrication charts or instructions shall be included in the manual. All lubricants, fluids, and associated products identified in the manual, supplemental data, or separate lubrication chart must have a Government identifier, (military specification number, NSN, and so forth) or manufacturer and part number which identifies the product beyond the product name and provides the user with requisitioning information.

3.4.12 Spare and repair parts. If required, the COTS manuals shall be supplemented with applicable spare and repair parts breakdown information. Supplemental data shall not duplicate data in the existing manual. The parts breakdown data shall be presented in contractor format and in the following order:

a. Introduction (Section I).
b. Parts list and illustrations (Section II).
c. Special support equipment list and illustrations (Section III).
d. Cross-reference indices (Section IV).

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3.4.12.1 Introduction - Section I. The introduction shall contain how-to-use information. The introduction shall explain the use of the parts equipment breakdown listings and the reference indices. The column headings, symbols, abbreviations, codings, notes, and other data used shall be explained in detail.

3.4.12.2 Parts list and illustrations - Section II. All spare and repair parts shall be listed by assembly. They shall be shown on the associated parts breakdown illustration. Unless otherwise specified in the TMCR (see 6.2), these assembly illustrations and parts lists shall be arranged in top-down breakdown (next higher assembly) sequence.

3.4.12.3 Illustrations. Illustrations shall be detailed line drawings and identify all repairable or replaceable parts. Each assembly illustration (or set of illustrations, if multi-sheets) shall precede the associated parts listing. Exploded views shall be provided when specified in the TMCR (see 6.2). Figures shall be numbered in ascending numerical sequence, and each item illustrated shall be assigned an item or index number. Item or index numbers shall be arranged in numerical clockwise sequence. Leader lines, with arrowheads, shall be drawn from each of the callouts (item or index no.) to the items.

3.4.12.4 Parts list. The parts list for the associated illustration shall bear the same title as the illustration and shall contain the columns listed:

a. Column (1), FIGURE-ITEM NO.
b. Column (2), SOURCE, MAINTENANCE AND RECOVERABILITY (SM&R) CODE.
c. Column (3), CAGE (Commercial and Government Entity) Code per H4/H8.
d. Column (4), PART NO. (actual manufacturers, vendors part, drawing, and military specification number).
e. Column (5), DESCRIPTION AND USABLE ON CODE (UOC) (item name and provisioning nomenclature (if required) and UOC (when more than one model or configuration is covered in the manual)).
f. Column (6), QTY (amount or number per figure or assembly)

3.4.12.5 Special support equipment - Section III. This section shall include special tools, test equipment, and special support equipment. Each tool or equipment shall be illustrated and a list prepared in the same format as for the parts list in section II. The basis of issue shall be stated for each kit or set on the last line of the Description and Usable on Code column. The basis of issue (BOI) provided by the Government, shall state the quantity per unit, division, company (CO), battery or other organizational structure or division, for example, BOI: 1 PER LETTERED CO. Items listed as part of a kit or set shall be shown as quantity per kit or set as applicable, and the BOI for the kit or set shall be stated for the kit or set line item entry.

3.4.12.6 Cross-reference indices - Section IV. Unless otherwise specified in the TMCR (see 6.2), this section shall include the following indices in the order presented:

a. Figure and item number index. This index, keyed on figure and item number in numerical sequence, shall cross-reference applicable CAGE Code number, part number, and the NSN.

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b. Part number index. This index shall key on part number in alphanumeric sequence and reference applicable CAGE Code, NSN, figure, and item number.
c. NSN index. This index shall list the NSN by ascending National item identification number, (the last nine digits). The NSN shall reference the applicable figure and item number.
d. Reference designator index. When figures or illustrations, such as electronic circuit diagrams, use reference designators in lieu of item numbers, an index listed by reference number in alpha numerical order shall cross-reference the applicable figure and item numbers.
e. Original equipment manufacturer (OEM) or true vendor. When applicable, the OEM part number shall be cross-referenced to true vendor part number. The part numbers shall be in alpha numeric sequence.

3.5 Accuracy. Essential corrections or changes to the manual shall be included in the supplemental data.

3.6 Changes and revisions to supplemental data. When the Government determines that changes or revisions are required to correct or update supplemental data, the data shall be changed or revised. Unless otherwise specified in the TMCR (see 6.2), changes and revisions shall be final reproducible copy and in the same style (including type style and point size) and format as the original supplemental data.

3.6.1 Change packages. When specified in the TMCR (see 6.2), change packages shall be in accordance with MIL-DTL-24784.

3.6.2 Revisions. When specified in the TMCR (see 6.2), revisions shall be in accordance with MIL-DTL-24784.

4. VERIFICATION

4.1 Quality assurance provisions. The quality assurance requirements for delivery of book plans, review draft copies, preliminary technical manuals, final reproducible copy, technical manuals, replenishment materials, changes and revisions shall be in accordance with MIL-DTL-24784.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but it is not mandatory.)

6.1 Intended use. COTS manuals and supplemental data are intended for support of COTS equipment used by the Department of Defense. (This specification is not intended to standardize the style, format, size, or content of COTS manuals, but to provide evaluation criteria for acceptance of COTS manuals and provide the requirements for supplemental data for selection by the Government.)

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification (or any TMCR referencing this specification).
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).
c. A copy of the COTS manual for Government reproduction and use is required to determine whether it is acceptable (Appendix A) for use or whether supplemental data is required (see 3.2).
d. When an existing COTS manual is reviewed, the acceptability certificate (Appendix B) should be provided (see 3.2).
e. Specification tailoring requirements for supplemental data by selection of content and format requirements (Appendix C) (see 3.4).
f. An ITPS is other than as specified (see 3.4.1).
g. Deliverable products as specified (see 3.4.2.2 and 3.4.2.3).
h. When supplemental data is required, contents should be specified (see 3.4.3).
i. If a copyright license has been granted to the COTS manual or data, obtain permission for Government reproduction and use (see 3.4.4).
j. Overprint the technical manual identification publication number and date , when specified (see 3.4.8 and 3.4.9).
k. Illustrated parts lists are other than as specified (see 3.4.12.2).
l. If exploded views are not required (see 3.4.12.3).
m. If cross reference indices are not required (see 3.4.12.6).
n. Changes to the supplemental manual should be specified only when they pertain to technical manual corrections resulting from a hardware change or modification. They may also be specified if it has been determined that significant omissions of technical data or information has been identified. Do not specify a change to correct minor or editorial corrections. When changes are justified on the above basis, the change package should include the correction of all outstanding temporary or interim changes and may include minor changes (see 3.6.1). The change package should be developed in accordance with MIL-DTL-24784.
o. When revisions are required (see 3.6.2)
p. Packaging requirements (see 5.1).

6.3 Technical manual acquisition. This specification (or a TMCR based on this specification) must be listed on the Contract Data Requirements List (DD Form 1423) in order to acquire the

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technical manuals described by this specification, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

6.3.1 Sample COTS manual. When COTS manuals are required for evaluation to determine their acceptability, the solicitation should state how many samples of the COTS manuals should be included in the bid package.

6.3.2 Sample qualification. Sample COTS manuals should be evaluated by use of Appendix A, Manuals Evaluation Checklist or other applicable document(s) determined by the Government. It should be noted that preestablished checklists may not contain all requirements, but each manual's content requirements depend on the item(s) covered by the manual. As well, the same concept applies when using Appendix C Content and Format Selection Summary or other equivalent contractual document(s) to solicit or contract for supplemental data to augment the acceptable COTS manual.

6.4 Definitions. The words or phrases used throughout this specification must be as defined in MIL-DTL-24784 and the following.

6.4.1 COTS manuals. Manuals available off-the-shelf from a commercial source which include operation, maintenance, and other instructions for commercial equipment. Commercial manuals are developed to support the equipment in the commercial market.

6.4.2 Supplemental data. Any data whether by change, addition, or separately numbered manual that augments or complements a COTS manual to make the manual acceptable for use by the Department of Defense.

6.5 Subject term (key word) listing.

Supplemental data

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

MANUALS EVALUATION (COTS)

A.1. SCOPE

A.1.1 Scope. This appendix may be used by contracting activities to evaluate COTS manuals to determine their acceptability for use by Department of Defense activities. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

A.1.2 Application. This appendix is intended to be copied or reproduced and completed to determine whether or not the reviewed manuals prove to be acceptable or non-acceptable. Appendix A, Manuals Evaluation Checklist and Appendix B, Acceptability Certification are companions in that decisions reflected on the completed certification reflect evaluations shown on the completed checklist.

A.1.3 Limitations. This appendix is limited in that it provides evaluation results and recommended actions(s) to be taken concerning the use and acquisition of COTS manuals only. In itself, it is not to be construed as an acquisition document, but may be helpful in selecting applicable supplemental data.

A.2. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

A.3. MANUAL EVALUATION CHECKLIST

A.3.1 Explanations of columns - Manuals Evaluation Checklist.

a. The REQUIREMENTS -(Paragraph) column contains the name of the requirement followed by the number of the paragraph where the requirement is located.
b. The ACCEPTABLE column has four subcolumns. If the manual contains the adjacent requirement and it is acceptable, mark an "X" in the YES column. If the manual does not contain the requirement or it is not acceptable, mark an "X" in the NO column. If the manual does not completely contain the requirements, or if the material is incorrect, mark an X in the INCOR/IMCOMP (INCORRECT/INCOMPLETE) column and make appropriate comments in the REMARKS column. If the listed requirement is not needed, mark an "X" in the NA (NOT APPLICABLE) column.
c. Make necessary comments in REMARKS column. If more space is needed, add extra sheet(s) of paper.

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APPENDIX A

MANUALS EVALUATION CHECKLIST
(Commercial Off-The-Shelf)

Proc/contr no: _____ Equipment model no: _____

Manual pub no: _____ Equipment name: _____

REQUIREMENTS - (PARAGRAPH)	ACCEPTABLE				REMARKS
(Make an "x" in applicable blocks. Fill in additional requirements, if applicable)	YES	NO	INCOR/ INCOMPL	NA	
Comprehensibility/readability - (3.2.1.1)					
Arrangement - (3.2.1.2)					
Legibility - (3.2.1.3)					
Size - (3.2.1.4)					
Safety - (3.2.1.5)					
Dangers, warnings/cautions/notes - (3.2.1.6)					
Federal protection standards - (3.2.1.7)					
Illustrations - (3.2.1.8)					
Maintenance - (3.2.1.9)					
Cover - (3.3.1.1)					
Table of contents - (3.3.1.2)					
Safety precautions - (3.3.1.3)					
Introduction - (3.3.2)					
Preparation for use - (3.3.3)					
Installation instructions - (3.3.4)					
General principles of operation - (3.3.5)					
Operating instructions - (3.3.6)					

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APPENDIX A

MANUALS EVALUATION CHECKLIST (Cont'd)
(Commercial Off-The-Shelf)

Proc/contr no: _____ Equipment model no: _____

Manual pub no: _____ Equipment name: _____

REQUIREMENTS - (PARAGRAPH)	ACCEPTABLE				REMARKS
(Make an "x" in applicable blocks. Fill in additional requirements, if applicable)	YES	NO	INCOR/ INCOMPL	NA	
Maint & servicing instruction (prevent & correct) - (3.3.7)					
Cleaning and lubrication - (3.3.7.1)					
Performance verification - (3.3.7.2)					
Inspection - (3.3.7.3)					
Troubleshooting - (3.3.7.4)					
Disassemble, repair, replace, reassemble - (3.3.7.5)					
Reprogramming - (3.3.8)					
Preparation for shipment - (3.3.9)					
Storage - (3.3.10)					
Parts list - (3.3.11)					
Operational & maintenance illustrations - (3.3.12)					
Diagrams - (3.3.12.1)					
Overhaul instructions -(3.3.13)					

APPLICABLE REQUIREMENTS HAVE BEEN EVALUATED AS INDICATED ABOVE

PUBLICATIONS REVIEW ACTIVITY: _____

_____

_____

ACTIVITY REVIEWER: _____ DATE: _____

(signature)

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APPENDIX B

ACCEPTABILITY CERTIFICATION (COTS MANUAL)

B.1. SCOPE

B.1.1 Scope. This appendix may be used as required by the Government to certify the acceptability or nonacceptability of COTS manuals. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

B.1.2 Application. This appendix is intended to be copied or reproduced and completed when the COTS manuals are evaluated by use of Appendix A. It will reflect the evaluations shown on the Manuals Evaluation Checklist, Appendix A.

B.2. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

B.3. ACCEPTABILITY

B.3.1 Certification. If the COTS manual is determined to be acceptable, the ACCEPTABLE block will be marked with an "X", and one of the subboxes must be marked to indicate any further action required. When the COTS manuals are not acceptable, the NOT ACCEPTABLE block will be marked.

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APPENDIX B

ACCEPTABILITY CERTIFICATION
(COTS MANUAL)

Procurement/Contract No: _____

Equipment Model No: _____

Equipment Name: _____

Manual Publication No: _____

☐ - ACCEPTABLE:

- ☐ No further action required - use as is.
- ☐ Make corrections/substitutions - (attach sheet specifying details).
- ☐ Supplement manual in accordance with Appendix C - use completed Evaluation Checklist to Content/Format/Selection Summary Sheet (Appendix C).

☐ - NOT ACCEPTABLE (Prepare TM per applicable military specification).

The above recommendation(s) are based on the completed Manual Evaluation Checklist.

PUBLICATIONS REVIEW ACTIVITY: _____

_____

_____

_____

ACTIVITY REVIEWER: _____
(signature)

DATE: _____

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APPENDIX C

CONTENT AND FORMAT SELECTION SUMMARY

C.1. SCOPE

C.1.1 Scope. Completion of the Content and Format Selection Summary Sheet may be used for acquisition of supplemental data for COTS manuals This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

C.2. APPLICABLE DOCUMENTS

(This section is not applicable to this appendix.)

C.3. SELECTION SUMMARY

C.3.1 Content and Format Selection Summary Sheet(s). The summary sheet(s) (the following pages) may be reproduced or adapted for specific application; however, the content and format should remain unchanged.

C.3.2 Completion of Summary Sheets. The Government is responsible for filling in the pertinent information (contract number, manufacturer, model number, acquiring or contracting activity, and so forth) at the top of the sheet and for indicating data required. All supplemental data required by this specification for a specific acquisition should be indicated by an "X" in the "Selected Content" column.

C.3.3 Explanation of columns - Content and Format Selection Summary Sheet. Column (1), self-explanatory. Columns (2) and (3) are the names of requirements and their respective paragraph numbers in this specification. Column (4a) should be marked with an "X" for each item or requirement applicable to the solicitation or acquisition. Column (4b) is provided to allow for remarks when explanations of selected requirements are needed.

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APPENDIX C

CONTENT/FORMAT SELECTION SUMMARY SHEET

SUPPLEMENTAL DATA FOR: _____

MANUFACTURER: _____

_____

_____

CONTRACT NO: _____

NOTE: Applicable requirements are indicated by an "x" in column 4a or explanation in column 4b.

(1) Item No.	(2) Requirements	(3) Applicable Paragraph No.	(4) Selection	
			(a) Selected Content	(b) Explanation/Remarks
1	Identifying technical publication sheet	3.4.1		
2	Style, format, form	3.4.2		
3	Safety precautions	3.4.3c		
4	Table of contents	3.4.3d		
5	Reporting of errors statement	3.4.3e		
6	Maintenance forms and records	3.4.3f		
7	Administrative storage	3.4.3g		
8	Destruction of military material	3.4.3h		
9	Lubrication instructions	3.4.3i 3.4.11		
10	Preventive maintenance	3.4.3j		
11	Spare/repair parts	3.4.3k 3.4.12		

Sheet 1 of _____

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APPENDIX C

CONTENT/FORMAT SELECTION SUMMARY SHEET (cont'd)

CONTRACT NO: _____

(1) Item No.	(2) Requirements	(3) Applicable Paragraph No.	(4) Selection	
			(a) Selected Content	(b) Explanation/Remarks
12	Recommend changes, User Activity Comment Sheet, TM deficiency report	3.4.3l		
13	Others as indicated on Appendix A Evaluation Checklist	3.4.3m		
14	Copyrights/proprietary information	3.4.4		
15	Warranty information	3.4.5		
16	Advertising	3.4.6		
17	Equipment/model coverage	3.4.7		
18	End item identification	3.4.8		
19	Publication number and date	3.4.9		
20	List of effective pages	3.4.10		
21	Accuracy	3.5		
22	Changes Packages	3.6.1		
23	Revisions	3.6.2		

Sheet 2 of____

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CONCLUDING MATERIAL

Review activity:
Navy - EC

Preparing activity:
Navy - SH
(Project TMSS-N282)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4,5,6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of this form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-24784/4A (SH)	2. DOCUMENT DATE *(YYMMDD)* 990315

3. DOCUMENT TITLE
COMMERCIAL OFF-THE SHELF (COTS) EQUIPMENT MANUAL REQUIREMENTS

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(if applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME Technical Point Of Contact (TPOC) Mr. Mickey Ander	b. TELEPHONE *(Include Area Code)* (1) Commercial (703) 602-8700 EXT 215 (2) AUTOVON
c. ADDRESS *(Include Zip Code)* Commander, Naval Sea Systems Command ATTN: SEA 0433, 2531 Jefferson Davis Hwy Arlington, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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